RESEARCH MEETING PREP · 24 SEP 2026

H2 BFS Experiment — Meeting Brief

Supervisors: Prof. Roni Stern · Prof. Rami Puzis · 196 queries · 9 algorithms

ACTION ITEMS — DELIVERY STATUS

1. Results table for all 196 queries

Full filterable table — 9 algorithms, GT rank per query, color-coded vs B1. Chips to filter: RAG failed, H1 improved, H1 worse.

`Tab: "All 196 Queries" in the dashboard`

2. Include reference documents that are not the GT

New analysis added: rows in deep-dives are now color-coded by LLM score. New "Top-1 soft retrieval" panel shows what each algorithm actually returned first and its LLM score. Summary tab shows that on average 16 non-GT docs score LLM=1.0 per query.

`Tab: "Deep Dives" (new panels) + "Summary" (soft context section)`

3. Three deep-dive query examples

Q118 (GT rank 68 for B2), Q37 (H3 soft-succeeds with LLM=1.0 non-GT), Q119 (B2 rank 38). Each shows: algorithm costs, top-1 soft quality, and full 25-doc table with LLM coloring.

`Tab: "Deep Dives"`

4. Sanity checks

4 checks: B1=RAG, H1→B1 as α→1, H2≡B1 (0/196 differences), H3 collapse note.

`Tab: "Sanity Checks"`

NUMBERS TO KNOW COLD

B1 TOP-1 (BASELINE)
70.4%
Mean rank 3.29 — best overall

H1 A=0.93 TOP-1
71.4%
+1pp vs B1, mean 3.35 (+0.06)

H2 VS B1 DIFFERENCES
0 / 196
Identical to B1 — tiebreaker never fires

H3 K=10 TOP-1
17.9%
−52.5pp vs B1 — centroid collapse

B2 DIST(C) TOP-1
14.8%
Worst — pure centroid ranking

NON-GT LLM=1.0 PER QUERY
~16
Avg across 196 queries. 29 docs at ≥0.8

Algorithm	Top-1 %	Top-3 %	Top-10 %	Mean rank
B1 dist(Q)	**70.4**	**83.2**	**91.8**	**3.29**
H1 α=.93	71.4	82.7	90.8	3.35
H1 α=.85	70.9	82.7	90.8	3.40
H1 α=.70	67.9	79.6	89.8	3.71
H2 lex	70.4	83.2	91.8	3.29
H3 k=10	17.9	43.9	91.8	5.05
H3 k=50	14.8	36.2	63.3	9.55
B2 dist(C)	14.8	36.2	62.8	9.87

H3 k=200 ≈ B2 — converges to pure centroid ranking at large k • Triangle: ∠Q≈45.7°, ∠C≈70.8°, ∠GT≈70.6° • ROC-AUC(LLM, d(GT))=0.9974

EXPECTED QUESTIONS — STRONG ANSWERS

Q1 Why does H2 give exactly the same results as B1?

Because cosine distances in **768-dimensional space** with **64-bit floats** produce no exact ties in practice. H2 sorts by dist(Q) first, then uses dist(C) as a tiebreaker — but in **196/196** queries, zero ties exist. The tiebreaker never fires. **This is actually a clean sanity check**: it confirms the pipeline is working correctly and that H2's design assumption (ties exist) is wrong for this data.

Q2 Why does the centroid hurt (B2 and H3 both collapse)?

The Q-C-GT triangle explains this. The centroid C is **not on the Q→GT axis**. Measured at Q, the angle to C is ∠Q≈45.7° and the angle to GT is ∠C≈70.8° — C and GT diverge ~46° as seen from Q. Ranking by dist(C) selects documents that are "central" to the pool on average, not documents that answer the specific query. **H3 k=10** restricts to the top-10 by dist(Q), then re-ranks by dist(C) — it effectively throws away the Q signal inside the shortlist.

Q3 Is the +1pp of H1 meaningful / statistically significant?

Honestly, it's marginal. **+1.0pp** on Top-1 (71.4% vs 70.4%) with N=196 is likely not significant. H1 also **loses slightly on Top-10** (90.8% vs 91.8%). The benefit is small and inconsistent across metrics. The main finding is that H1 cannot substantially beat B1 because the centroid is not a reliable signal for most queries.

Q4 What does the "soft retrieval" analysis show?

For every query, the pool contains on average **16 non-GT documents with LLM score 1.0** and 29 with score ≥0.8. So the binary "found GT?" metric is strict. The three deep-dive examples show two distinct failure modes: in Q118,

H3's top-1 pick has LLM=0.2 (genuinely irrelevant — the centroid led it in the wrong direction). In Q37, H3's top-1 pick has LLM=1.0 (a different valid answer — H3 softly succeeded despite missing the official GT).

Q5 What is the centroid exactly? How was it computed?

The centroid C for each query is the vector average (mean) of the top-k candidate documents' embeddings retrieved by the initial RAG pass. It represents the "center of mass" of the candidate pool. The near-universal Q-C-GT triangle structure (isosceles, nearly constant angles across all 196 queries) shows this centroid is geometrically consistent — but geometrically consistent doesn't mean useful for retrieval.

Q6 Why does H3 recover at Top-10 (91.8%)?

Because H3 only re-ranks within the top-10 by dist(Q) — it doesn't remove documents from the top-10 pool. If GT was in the top-10 by dist(Q) (which happens 91.8% of the time), it stays in the top-10 even after H3 re-ranks. H3 just moves GT's rank around within those 10 positions. At k=50 and k=200, the pool expands and can push GT further out, so Top-10 drops.

HANDLE CAREFULLY

! "Why didn't you test α values below 0.7?"

Good answer: at α=0.70 the centroid weight (0.30) already causes a **−2.5pp** drop. Below 0.7 the centroid increasingly dominates and performance converges toward B2 (14.8%). The trend is monotone and clear — further testing would not add new insight. **Frame this as a deliberate stopping point, not a gap.**

! "16 non-GT docs at LLM=1.0 — does this mean your GT labels are noisy?"

No — the GT is the single annotated answer from the dataset. Multiple documents can be equally valid answers to a question (especially in biomedical QA). The 16 other LLM=1.0 docs are genuinely relevant but were not the annotated GT. **This is a property of the dataset, not a labeling problem.** It does suggest that future evaluation should consider soft metrics (MRR over relevant docs, not just GT recall).

? "What's the next step — what does this suggest for future work?"

The centroid of the raw candidate pool is not a reliable signal. What might help: **(a)** an LLM-filtered centroid — average only the high-score docs instead of the whole pool; **(b)** iterative refinement — use the top-1 retrieved doc to update the query vector; **(c)** learned α per query — the optimal weighting is query-specific (some queries benefit from C, most don't). **Be concrete, name at least one of these if asked.**

? "Your Top-10 numbers — B1 and H3 k=10 both show 91.8%. Is that a coincidence?"

No, it's structural. H3 k=10 restricts its candidate set to exactly the top-10 by dist(Q), so its Top-10 *must* equal B1's Top-10: if GT is in B1's top-10 (91.8% of queries), it's still in H3's input set by definition. **This is a principled observation, not a coincidence — it confirms the code is correct.**

Overall narrative to lead with: All four meeting tasks are complete. The key finding is that centroid-based heuristics — whether as a tiebreaker (H2), a hybrid weight (H1), or a re-ranker (H3) — fail to meaningfully improve over standard dist(Q) retrieval. H2 adds nothing. H1 adds at most +1pp. H3 is actively harmful. The geometric analysis explains why: the pool centroid is a stable anchor in embedding space but it does not point toward the query's specific answer.